

5. Optimizing Application Deployment with Blue-Green and Canary Releases on Kubernetes

Problem Statement:

Rolling out new features or updates can be risky, especially in production environments. Blue-Green and Canary releases are deployment strategies that reduce the risk by enabling phased rollouts and quick rollbacks if issues arise.

Project Goals:

- Implement Blue-Green and Canary deployment strategies using Kubernetes.
- Automate the process of switching between environments to minimize downtime.
- Monitor and analyze the impact of new releases to ensure stability before full rollout.
- Establish rollback mechanisms to revert changes quickly in case of issues.

Tools Used:

- Kubernetes for container orchestration
- Argo Rollouts or Flagger for advanced deployment strategies
- Helm for Kubernetes package management
- Prometheus and Grafana for monitoring and performance analysis
- Istio or Linkerd for service mesh and traffic management
- Jenkins or GitLab CI/CD for pipeline automation

Evaluation Criteria for Deliverables, Presentation and Viva

Documentation	15.00%
Implementation	75.00%
Cost Optimization	10.00%